HIKET — next session

The level/trend discrepancy: what it is, and the hypotheses for it

7 August 2026 — working document, not for circulation

Abstract

Companion to `M&M_parameterization_working_document.pdf`. That document states the parameterisation assumptions and their evidence. This one states the problem that survives them. After correcting the error model and the pre-run anchor, the six models can reproduce the observed soil carbon *stock* but not its *rate of change*: the level and the trend make opposing demands, and no setting of the initialisation satisfies both. Six candidate explanations are recorded here with their status and the test that would decide each. None is ruled out except one, which was implemented and failed. A seventh issue — the observed rate itself is not yet uniquely defined — is listed first because every other gap is measured in its units.

1 Run in progress

Roihu jobs **509638–509643** launched 2026-08-07 ≈17:00 (SP1 509638, TP2 509639, TP3 509640, Yasso07 509641, Yasso15 509642, Yasso20 509643), commit `e7e56bb`, ≈13–19 h. First run with a correctly specified error model, the common pre-run anchor, and the ratio prior at 0.90. Pre-flight passed ($K = 200$, full N, blow-up rates identical to the pre-change baseline).

Expected: levels correct across all six; **Yasso15/20 still reporting a source**. The single most informative output is where σ_{init} lands under the corrected anchor at production length — estimated ≈1.33, i.e. still inverted, but that is extrapolated from 6000-iteration widths and could be wrong.

2 The discrepancy

Sweeping the 1917 anchor at posterior-median parameters, the level and the trend cannot be satisfied together:

Table 1: σ_{init} sweep. Observed: 1985 = 61.6, 2024 = 73.8, trend $+0.312\,\mathrm{tC\,ha^{-1}\,yr^{-1}}$.

	TP2		Yasso20	
σ_{init}	1985	trend	1985	trend
0.25	26.8	+0.299	23.3	+0.194
0.70	**60.3**	+0.137	42.4	+0.082
1.10	90.2	−0.008	**59.4**	−0.018

Get the level right and the trend is a third to a tenth of observed; get the trend right and the 1985 stock is less than half of observed. This is the problem to explain.

3 H6 first: the target is not yet defined

Every gap below is quoted against an observed rate that varies with two choices not yet made explicitly.

Table 2: Observed paired rate of SOC stock change ($\mathrm{tC\,ha^{-1}\,yr^{-1}}$), plot set defined by the data alone.

period	n	mean	10 % trimmed	median
1985–2006	348	+0.458	+0.409	+0.410
2006–2024	409	+0.209	+0.105	+0.071
1985–2024	316	+0.312	+0.260	+0.229

H6 — the observed rate is ambiguous by a factor of ∼3
Statistic: 1985–2024 spans +0.229 (median) to +0.312 (mean), a 36 % range; 2006–2024 spans a factor of three. Per-plot changes are strongly right-skewed.
Plot set: a model-dependent subset (plots where the forward run succeeded) gave +0.116 for 2006–2024 where the data-defined set gives +0.209. A target must not depend on which plots a model managed to run.

Test / action: fix one plot set and one statistic, applied identically to observed and modelled rates. For an inventory the **mean** is defensible — total stock change is what is reported, and the median discards the high-accumulation plots carrying the sink — but the skew should be reported alongside: a mean–median gap this large is itself a result about the heterogeneity of accumulation.

Why first: the target spans ≈ +0.07 to +0.31, the same magnitude as the deficit being explained. Some of the discrepancy may be a choice rather than a mechanism.

4 The hypotheses

H1 — kinetics too slow to track the rising litter
The cascade models have an effective response time $\tau \approx 195$ yr and close only 18 % of any gap over 39 years; the observations imply $\tau \approx 33$–38. Yasso is fast enough (17–38 yr) but its rates are fixed by design, and its free transfer fractions only redistribute carbon among pools that all turn over within a decade — the one slow reservoir (H, $\tau \approx 855$ yr) has a *fixed* share. So Yasso's response time is close to a design constant.
Status: identified. Freeing the slow-rate prior (SD 0.15 → 0.50) made it *worse* — α_H drifted down 0.0053 → 0.0031 and the trend moved only 0.116 → 0.130 — because the likelihood does not reward the trend (below).
Test: make the likelihood target the stock change, then see where the rates go. A perfect trend fit is currently worth 3.4 nats against a parameter excursion costing 8.9; adding the paired stock change as an explicit observation with its own SE (0.035) would be worth ≈ 13, which flips the balance.

H2 — the litter trend is understated
σ_{input} absorbs any error in the litter *level*, so only an understated *rate of increase* causes under-accumulation. The product rises ×1.35 over 1986–2024 against a growing stock rising ×1.42.
Status: **untested**, and the only remaining input-side hypothesis that is testable with data in hand.
Test: sweep a multiplier on the litter trend (holding the level) and find what rise would be required; check it against the growing-stock rise and the fitted elasticity. Same shape as the σ_{init} sweep and equally decisive.

H3 — the 1985 campaign reads low
VMI8 samples 0–5 and 5–20 cm where the later campaigns sample 0–10 and 10–20 cm, and it could not be validated against official LUKE figures. If it reads low, the observed trend is overstated.
Status: plausible and **not resolvable with these three campaigns.** The obvious test — compare over 2006–2024, which excludes 1985 — has almost no power: the total change there is $+3.8\,\mathrm{tC\,ha^{-1}}$ on a stock of $\sim$70 (3.2σ), so a model capturing 50 % of the trend is statistically indistinguishable from one capturing 100 %. Essentially all the discriminating information sits in the 1985–2006 leg ($+9.6\,\mathrm{tC\,ha^{-1}}$, 9.5σ) — the contested one.
Consequence: H1 and H3 predict the same observable. Record as a limit of the data, and as a stated sensitivity, rather than as an open analysis task.
Exception: Yasso20 predicts -0.147 against $+0.209$ over the trusted campaigns, a gap of 5.5 SE. Its failure stands regardless of 1985.

H4 — a missing input component with its own trend
Understorey litter is absent from the product entirely, and its share plausibly changed with canopy closure over the period. A missing component with a rising trend would look exactly like H2.
Status: untested; blocked on the LUKE MUSTIKKA ground-vegetation data, which is not held locally. Relevant that two models put σ_{input} *below* 1, against a C4b prior centred at 1.30 precisely because understorey is missing.
Test: would appear as a productivity-dependent trend deficit, since understorey is proportionally largest where tree litter is smallest.

H5 — a missing process: stabilisation capacity
Clay adds +0.050 to the residual R^2 beyond litter ($n = 456$, $p < 10^{-7}$), with the right sign: less over-prediction on higher-clay soils. No model in the set has any edaphic control on decomposition — a shared blind spot.
Status: evidence in hand, mechanism not implemented. Realistically the second paper.
Caveat: CEC adds more (+0.087) but is measured on the same samples as SOC and organic matter drives CEC — circularity. Use clay.

H0 — REJECTED: historical depletion of the 1917 soil
That early-20th-century management (litter raking, forest grazing, slash-and-burn) removed organic matter before it entered the soil, leaving 1917 stocks below what a biomass proxy implies.
Status: **implemented and rejected** (§2). The anchor sweep shows the level and the trend make opposing demands: no value of σ_{init} satisfies both, in either family. The hypothesis may remain true as history, but it is not the mechanism behind the trend deficit.
What survives: initialisation controls the sink *sign* — Yasso20 spans -0.09 to $+0.23$ across the sweep — but not its magnitude.

5 Suggested order

1. **H6** — fix the target. Cheap, and it may shrink the discrepancy before anything is attributed to a mechanism. Everything else is measured in its units.

2. **H2** — the litter-trend sweep. The only remaining testable input hypothesis, and the same shape as the sweep already run.

3. **H1** — change what the likelihood targets, then re-read the rates. Expensive, and its answer depends on the target settled in (1).

4. **H3** — write up as a limit of the data with a stated sensitivity.

5. **H4, H5** — second paper, or as discussion; H4 needs external data.

These are not exclusive. Several could contribute simultaneously, and the ordering above is by cost and dependency, not by expected importance. In particular H1 and H3 are observationally degenerate over the available campaigns, so neither can be confirmed by ruling the other out.